

Introduction to Quantum Computing

Key Images: Week 2

What Makes a Good Qubit Modality? > Requirements for the Physical Implementation of Quantum Computers

DIVINCENZO CRITERIA

REQUIREMENTS FOR THE PHYSICAL IMPLEMENTATION OF QUANTUM COMPUTATION	
D1: Scalable qubits	Scalable physical system of well-defined, characterized qubits
D2: Initialization	Prepare a simple, fiducial input state
D3: Measurement	Measure the qubit state
D4: Universal gate set	Perform a universal set of gate operations with high fidelity
D5: Coherence	Robustly represent quantum information (long coherence times)
REQUIREMENTS FOR ROUTING QUANTUM INFORMATION	
D6: Interconversion	Ability to interconvert stationary and flying qubits
D7: Communication	Ability to transmit flying qubits faithfully between two locations

Qubit Modalities > State of the Art

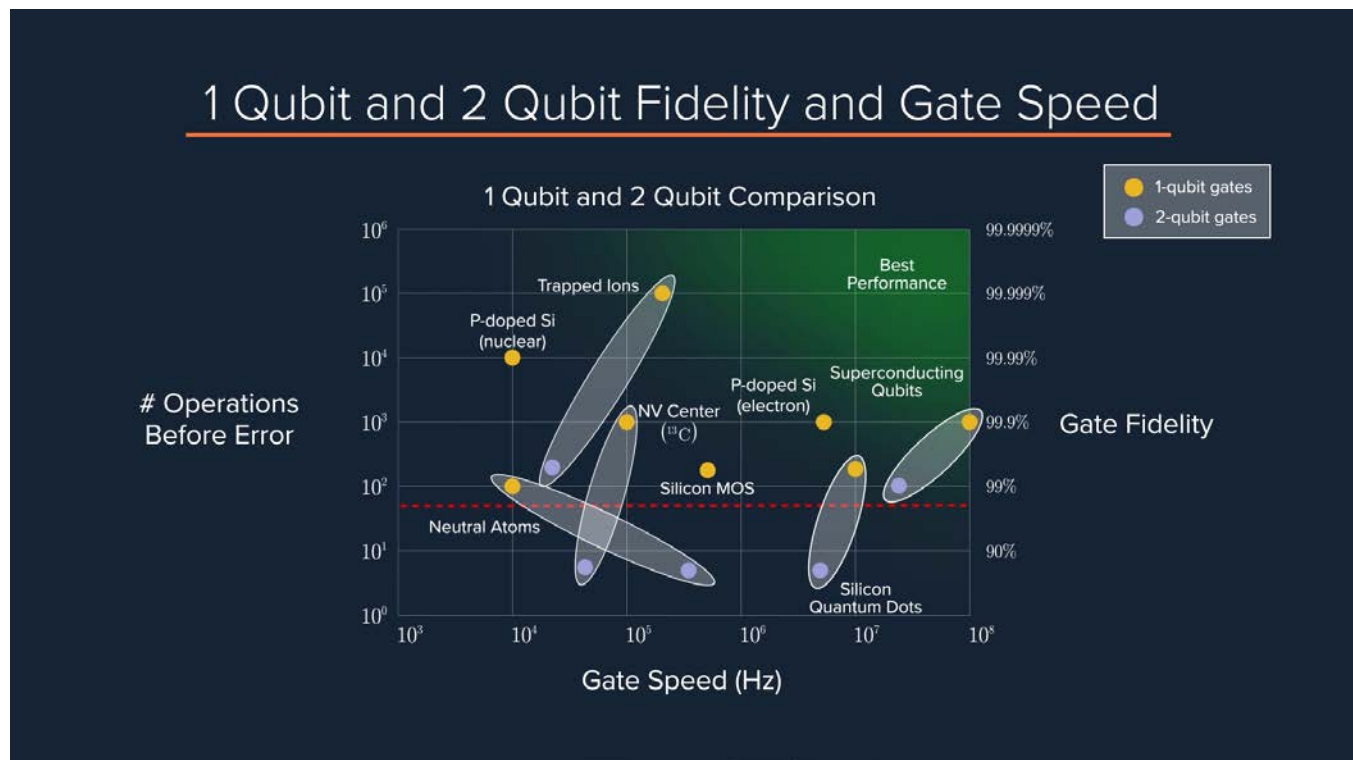
	D1 SCALABLE SYSTEM	D2 INITIALIZATION	D3 MEASUREMENT	D4 UNIVERSAL GATES	D5 COHERENCE	D6 INTERCONVERSION	D7 COMMUNICATION
SiGe Quantum Dots							
Doped Silicon							
NV Centers							
Neutral Atoms							
Trapped Ions							
Superconductors							

■ Sufficient demonstrations exist to proceed to the 100 qubit level

■ Concepts and/or first demonstrations exist

■ No realistic concepts yet developed

Qubit Modalities > Comparing Qubit Modalities



Qubit Modalities > Comparing Qubit Modalities

		SUPER-CONDUCTING	TRAPPED IONS	SILICON QUANTUM DOTS		IMPURITY-DOPED SILICON		IMPURITY CENTERS	TRAPPED NEUTRAL ATOMS
QUBIT MODALITY	MATERIALS	Al	Yb+, Ca+, Sr+, Be+, Ba+, Mg+	isotopically purified Si, SiGe	isotopically purified Si	Phosphorus doped isotopically purified silicon		Nitrogen vacancies in diamond	Rb, Cs, Ho
	TYPE	transmon	hyperfine & optical transitions	quantum dot 2DEG	quantum dot Si MOS	implanted dopant		nuclear + electron spin	Rydberg states
	CONTROL/RO	microwaves	microwaves & optics	baseband or microwave	baseband or microwave	microwaves		microwaves & optics	microwaves & optics
	STATE	junction phase	atomic state of electron	electron spin	electron spin	electron spin	nuclear spin	nuclear + electron spin	Rydberg state
STATE-OF-ART TIMES (ns)	T1	50,000	>1E14 (years)	1,000,000,000	2,000,000,000	5,000,000,000	>1E14	100,000,000	5,000,000,000
	T2	100,000	50,000,000,000	400,000	1,200,000	100,000	1,000,000,000	20,000,000	1,000,000,000
	1QB GATE	10	0.05 / 2,000	100	2,000	200	100,000	10,000	100,000
	2QB GATE	40	1,600 / 30,000	200	600	1.6	TBD	25,000	3,000
	RO	200	30,000	1,000	100,000	1,000,000	50,000,000	50,000	150,000,000
FIDELITY	1QB	99.90%	99% / 99.997%	99.60%	99.5%	99.90%	99.99%	99.90%	99.9%
	2QB	99.0%	99.8% / 99.92%	80%	94.7%	86.7%	TBD	82%	80%
	RO	99%	99.99%	99%	95%	94.00%	99.90%	94%	99.90%
CLOCK (MHz)	1QB GATE	100.00	20,000 / 0.50	10.00	0.50	5.00	0.01	0.10	0.01
	2QB GATE	25.00	0.60 / 0.03	5.00	1.00	625	TBD	0.04	0.33
	READOUT	5.00	0.03	1.00	0.01	0.001	0.00002	0.02	0.006